

Week 9 Applied Class Worksheet Solutions

COMP2048

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Task 1. *Big-O Practice*

For each of the following statements, determine whether it is true or false. Briefly justify your answer.

- (a) $7n^2 + 3n + 100 = O(n^2)$.
- (b) $n^2 = O(n \log n)$.
- (c) $n \log n = O(n^2)$.
- (d) $2^n = O(n^{100})$.
- (e) $n^{100} = O(1.0001^n)$.

Solution.

- (a) True. Highest order term is n^2 .
- (b) False. n grows faster than $\log n$.
- (c) True. n grows faster than $\log n$.
- (d) False. Exponential functions grow faster than polynomial functions.
- (e) True. Any exponential function a^n with $a > 1$ eventually dominates any polynomial.

Task 2. *Time Complexity Classes*

For each of the following statements, determine whether it is true or false. Briefly justify your answer.

- (a) If $L \in \text{TIME}(n^2)$, then $L \in \text{TIME}(n^3)$.
- (b) If $L \in \text{TIME}(n^3)$, then $L \in \text{TIME}(n^2)$.
- (c) If $L \in \text{TIME}(2^n)$, then $L \in \text{P}$.

- (d) If $L \in \text{TIME}(2^n)$, then $L \notin P$.
- (e) If $L \in \text{TIME}(n^{10^6})$, then $L \in P$.
- (f) If $L \in P$, then there exists some constant k such that $L \in \text{TIME}(n^k)$.
- (g) Some problems in P are undecidable.
- (h) $\text{HALT}_{\text{TM}} \in \text{TIME}(2^n)$.

Solution.

- (a) True. If $L \in \text{TIME}(n^2)$, then there is a deterministic Turing machine deciding L in time $O(n^2)$. Since $n^2 = O(n^3)$, the same machine also runs in time $O(n^3)$. Hence

$$L \in \text{TIME}(n^3).$$

- (b) False. Knowing that a language has an $O(n^3)$ -time decider does not imply that it has an $O(n^2)$ -time decider.
- (c) False. A language being decidable in exponential time does not necessarily mean it is decidable in polynomial time.
- (d) False. Some languages in $\text{TIME}(2^n)$ are also in P .
- (e) True. The function n^{10^6} is still a polynomial, since 10^6 is a constant. Therefore, if

$$L \in \text{TIME}(n^{10^6}),$$

then L is decidable in polynomial time, so $L \in P$.

- (f) True. This is exactly the definition of P :

$$P = \bigcup_k \text{TIME}(n^k).$$

So if $L \in P$, then there exists some constant k such that

$$L \in \text{TIME}(n^k).$$

- (g) False. Every language in P is decidable by definition. A language in P has a deterministic Turing machine that decides it in polynomial time, so it cannot be undecidable.
- (h) False. The language HALT_{TM} is undecidable. Every language in $\text{TIME}(2^n)$ is decidable, since it is decided by a Turing machine that halts on every input within $O(2^n)$ steps. Therefore,

$$\text{HALT}_{\text{TM}} \notin \text{TIME}(2^n).$$

Task 3. *A Linear-Time Two-Tape Decider*

Let

$$L = \{0^k 1^k \mid k \geq 0\}.$$

- (a) Construct a deterministic two-tape Turing machine that decides L in $O(n)$ time. You may describe the machine at a high level.
- (b) Explain why this does not contradict the fact that the usual one-tape marking algorithm we saw in the lecture for L takes $O(n^2)$ time.

Solution.

- (a) We describe a deterministic two-tape Turing machine deciding

$$L = \{0^k 1^k \mid k \geq 0\}.$$

On input w , the machine works as follows.

First, it scans the initial block of 0's on Tape 1. For each 0 it sees, it writes one 0 on Tape 2. Thus Tape 2 stores the number of 0's seen.

Once the machine reaches the first 1 on Tape 1, it begins matching the 1's against the copied 0's on Tape 2. For each 1 read on Tape 1, the machine moves left by one copied 0 on Tape 2.

If the machine ever sees a 0 after it has started reading 1's, it rejects. It also rejects if the number of copied 0's and the number of 1's do not match. Otherwise, it accepts.

The machine decides L because an input is accepted exactly when it consists of some number of 0's followed by the same number of 1's.

The running time is $O(n)$. The machine scans Tape 1 once from left to right. Tape 2 moves right while copying the 0's and then moves left while matching them with the 1's. Thus the total number of steps is at most a constant times the input length n .

Therefore L is decidable by a deterministic two-tape Turing machine in $O(n)$ time.

- (b) This does not contradict the $O(n^2)$ one-tape algorithm from lecture. On a one-tape Turing machine, the head repeatedly scans back and forth across the input to match each 0 with a 1, giving $O(n^2)$ time.

On a two-tape Turing machine, the second tape stores the number of 0's, so the machine avoids these repeated scans. This shows that exact running times such as $O(n)$ and $O(n^2)$ can depend on the machine model. However, both bounds are polynomial, so both algorithms show that

$$L \in P.$$

Task 4. *Regular Languages Are in P*

Prove that every regular language is in P.

Solution.

We have already shown that Turing machines can simulate DFA by scanning the input from left to right and making a decision after reading the last symbol. On an input of length n , the Turing machine will make a decision after $O(n)$ steps. Therefore, all regular languages are in $\text{TIME}(n) \subset \text{P}$.